MATH1307 Assignment 1

Code

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Student Details

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Introduction

This report investigates the relationship between the ASX All Ordinaries (ASX) Price Index and selected economic variables (Gold, Oil, and Copper) using time series analysis. The primary objective is to examine the time series objects and identify a suitable distributed lag model for ASX index.

The analysis involves four main steps: preparing and cleaning the dataset, assessing stationarity and transforming non-stationary variables, fitting different distributed lag models, and selecting the most appropriate model based on goodness-of-fit, residual diagnostics, and multicollinearity. By systematically comparing finite distributed lag, polynomial distributed lag, Koyck, and autoregressive distributed lag models, the report aims to provide a reliable framework for understanding and analysing the variables.

Load Packages

Hide

```
# Loading necessary packages
library(tseries)
library(urca)
library(forecast)
library(TSA)
library(x12)
library(car)
library(dLagM)
```

Data Preparation

- This step aims to gain an understanding of the dataset and its characteristics. It involves examining the data structure, performing the necessary cleaning, and converting variables into appropriate data types, thereby ensuring that the dataset is properly prepared for the subsequent stages of analysis.
- Initially, the Copper and Gold price variables were stored as character types. These were converted to numeric values. After this conversion, all variables were successfully formatted as time series.

Hide

```
#Read the data
setwd("C:/Users/tally/OneDrive/Desktop/Forecasting")
data <- read.csv("ASX_data.csv")
```

Hide

```
# Assess the dataset
head(data)
```

	ASX.price	Gold.price	Crude.Oil..Brent._USD.bbl	Copper_USD.tonne
	<dbl>	<chr>	<dbl>	<chr>
1	2935.4	611.9	31.29	1,650
2	2778.4	603.3	32.65	1,682
3	2848.6	565.7	30.34	1,656
4	2970.9	538.6	25.02	1,588
5	2979.8	549.4	25.81	1,651
6	2999.7	535.9	27.55	1,685

6 rows

Hide

```
str(data)
```

```
'data.frame':   161 obs. of  4 variables:
 $ ASX.price               : num  2935 2778 2849 2971 2980 ...
 $ Gold.price              : chr  "611.9" "603.3" "565.7" "538.6" ...
 $ Crude.Oil..Brent._USD.bbl: num  31.3 32.6 30.3 25 25.8 ...
 $ Copper_USD.tonne        : chr  "1,650" "1,682" "1,656" "1,588" ...
```

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```
# Convert to numeric values
data$Copper_USD.tonne <- as.numeric(gsub(",", "", data$Copper_USD.tonne))
data$Gold.price <- as.numeric(gsub(",", "", data$Gold.price))
str(data)
```

```
'data.frame':   161 obs. of  4 variables:
 $ ASX.price               : num  2935 2778 2849 2971 2980 ...
 $ Gold.price              : num  612 603 566 539 549 ...
 $ Crude.Oil..Brent._USD.bbl: num  31.3 32.6 30.3 25 25.8 ...
 $ Copper_USD.tonne        : num  1650 1682 1656 1588 1651 ...
```

Hide

```
# Convert to time series object
ASX_ts <-ts(data$ASX.price, start = c(2004,1), frequency = 12)
Gold_ts <-ts(data$Gold.price, start = c(2004,1), frequency = 12)
Oil_ts <-ts(data$Crude.Oil..Brent._USD.bbl, start = c(2004,1), frequency=12)
Copper_ts <-ts(data$Copper_USD.tonne, start = c(2004,1), frequency = 12)
data_ts <- ts(data,start = c(2004,1), frequency = 12)
```

Nonstationary analysis

- A non-stationary time series is one whose statistical properties, such as mean and variance, change over time (Chand, 2023). Non-stationarity is a critical issue because it can lead to misleading regression results due to spurious correlations. To assess stationarity, two approaches were used: visual inspection and formal unit root tests.
- The first approach involves visual inspection of the time series plots and their autocorrelation functions (ACF). These visual tools help identify potential trends and persistence in the data, providing an intuitive understanding of how each series evolves over time.
- The second approach uses the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) unit root tests, which are standard statistical procedures for formally testing the presence of a unit root and confirming non-stationarity. Combining these two methods allows for a thorough assessment of the data.

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```
# Function for plots
plot_series <- function(ts_data, name) {
  plot(ts_data, main=paste(name, "Time Series"), ylab="Value", xlab="Time", type="o") #time seri
es plot
  text(time(ts_data), ts_data,
     labels = substring(month.abb[cycle(ts_data)], 1, 1),
     pos = 3, cex = 0.7, col = rainbow(12)[cycle(ts_data)]) #showing seasonal patterns
  par(mfrow=c(2,2))
  acf(ts_data, main=paste("ACF:", name),cex.main=0.65) #ACF plot
  pacf(ts_data, main=paste("PACF:", name),cex.main=0.5) #PACF plot
  par(mfrow=c(1,1))
}

# Function for unit root test
unit_root_test <- function(ts_data,name) {
  adf_result <- ur.df(ts_data, type = "none", lags = 1, selectlags = "AIC") #ADF test
  print(summary(adf_result))
  pp_result <- ur.pp(ts_data, type = "Z-tau", lags = "short") #PP test
  print(summary(pp_result))
}
```

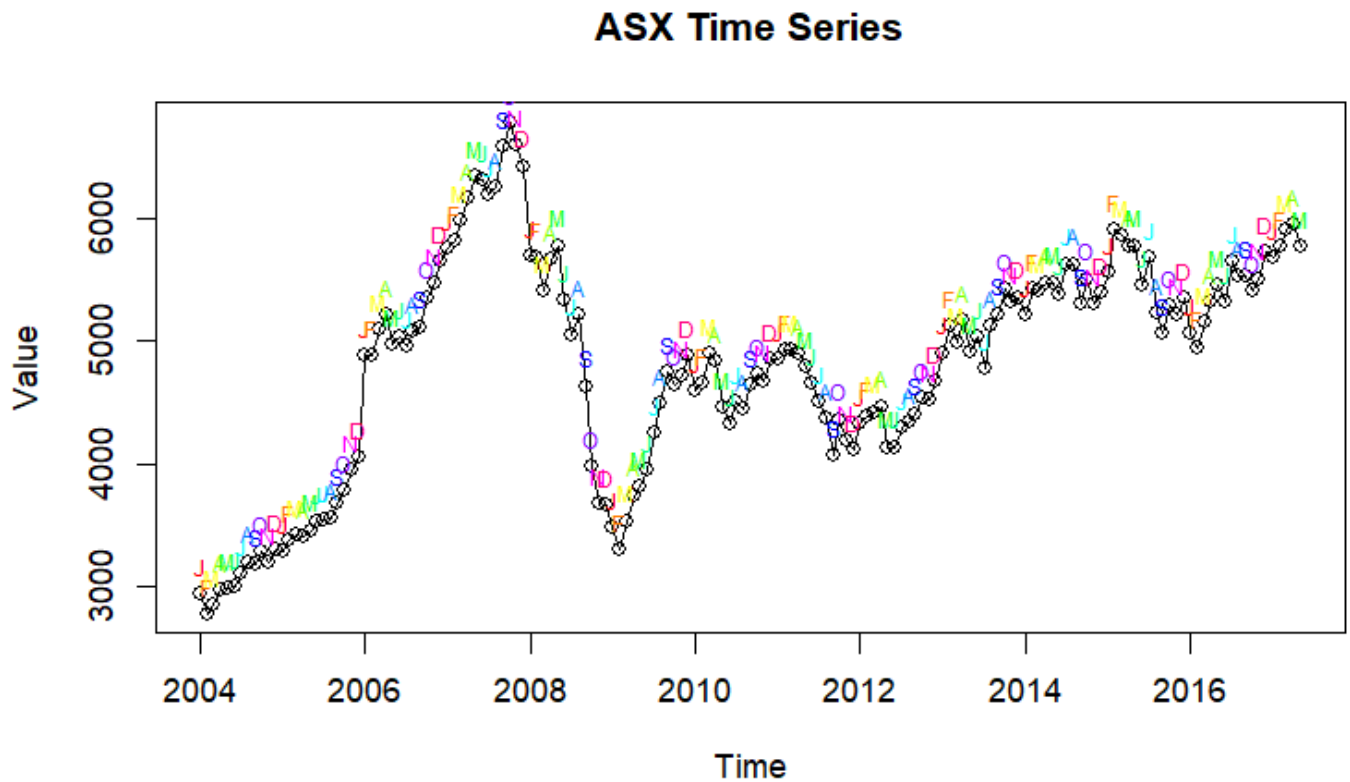
ASX variable

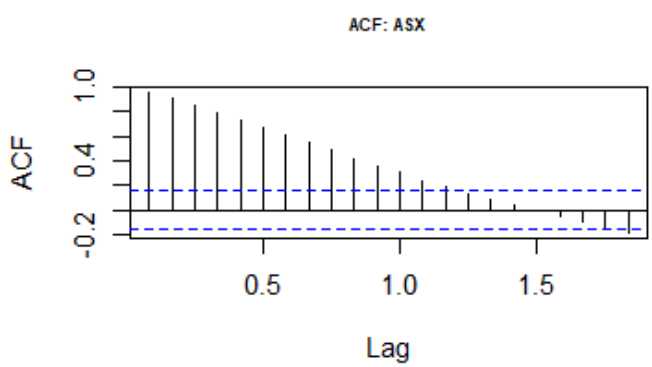
The ASX series shows a clear upward trend, indicating non-stationarity in the mean. The ACF exhibits a slow decay, and the PACF shows a high first autocorrelation. Formal testing using the ADF and PP tests confirms this observation: both test statistics exceed their respective critical values, indicating that the null hypothesis of a unit root is not rejected.

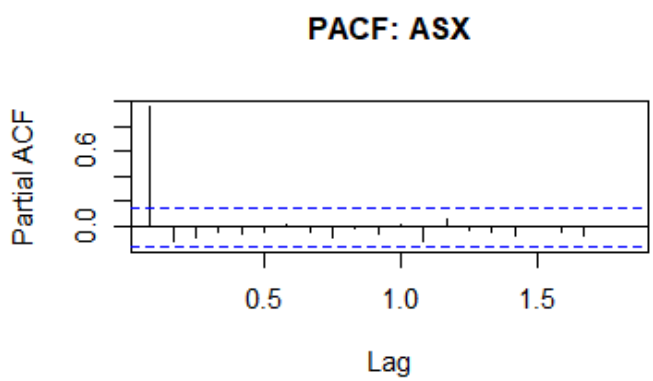
=> **Nonstationary**

Hide

```
# Running functions for ASX variable
plot_series(ASX_ts, "ASX")
```







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```
unit_root_test(ASX_ts, "ASX")
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression none


Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max 
-722.83 -103.12   36.63  138.79  809.17 

Coefficients:
           Estimate Std. Error t value Pr(>|t|)
z.lag.1    0.002107   0.003309   0.637    0.525
z.diff.lag 0.085167   0.079885   1.066    0.288

Residual standard error: 203.3 on 157 degrees of freedom
Multiple R-squared:  0.01074,	Adjusted R-squared:  -0.001866 
F-statistic: 0.8519 on 2 and 157 DF,  p-value: 0.4286


Value of test-statistic is: 0.6366 

Critical values for test statistics: 
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62


##################################
# Phillips-Perron Unit Root Test #
##################################

Test regression with intercept


Call:
lm(formula = y ~ y.l1)

Residuals:
    Min      1Q  Median      3Q     Max 
-683.35 -104.89   15.86  135.94  782.18 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 191.38104   87.09761   2.197   0.0295 *  
y.l1          0.96386    0.01781  54.107   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 200.7 on 158 degrees of freedom
Multiple R-squared:  0.9488,    Adjusted R-squared:  0.9485
F-statistic:  2928 on 1 and 158 DF,  p-value: < 2.2e-16


Value of test-statistic, type: Z-tau  is: -2.2025

         aux. Z statistics
Z-tau-mu           2.3366

Critical values for Z statistics:
                     1pct      5pct     10pct
critical values -3.472136 -2.879539 -2.576262
```

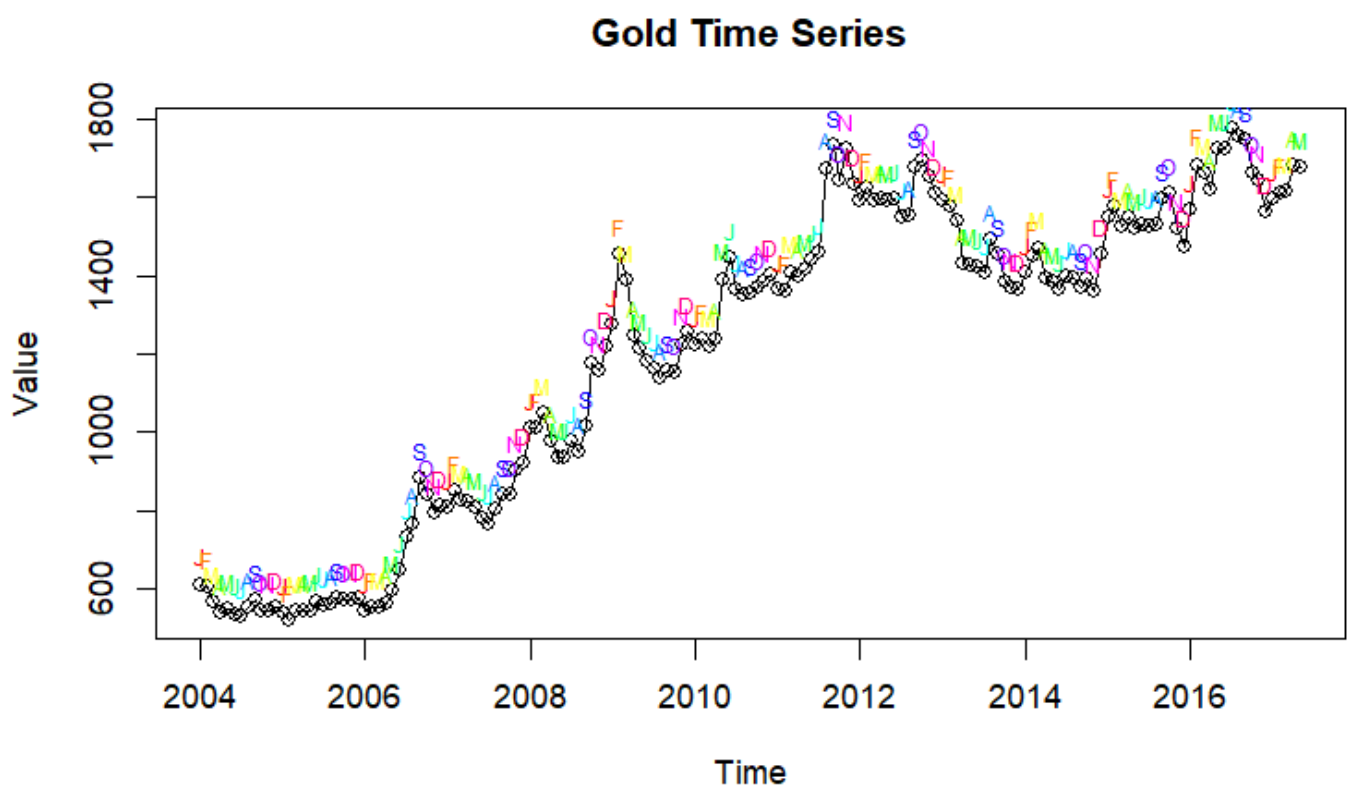
Gold variable

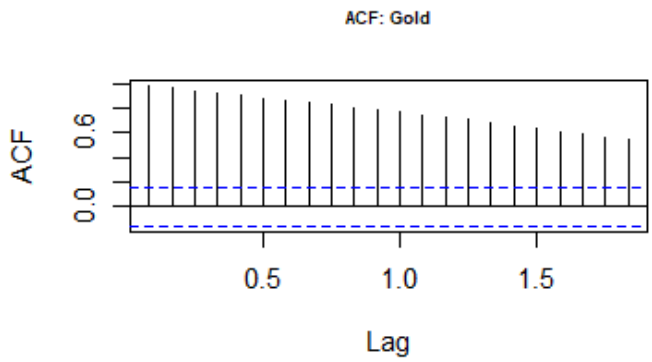
Gold prices display an upward trend from 2006 to 2012. The ACF decays slowly, and the PACF shows high initial autocorrelation. Formal testing supports this conclusion: the ADF test statistic is more negative than the critical value, and the PP test statistic exceeds the 5% critical value. Both tests fail to reject the null hypothesis of a unit root.

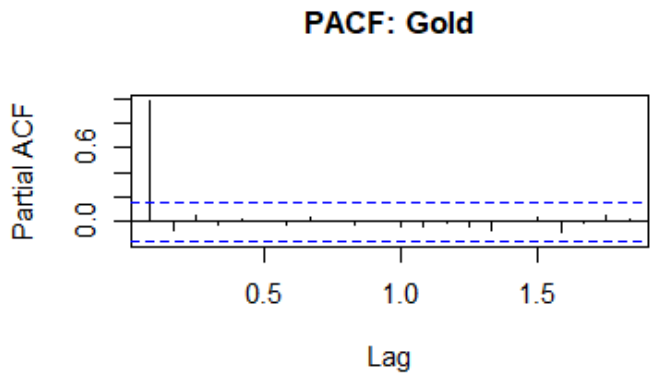
=> **Nonstationary**

Hide

```
# Running functions for Gold variable
plot_series(Gold_ts, "Gold")
```







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```
Gold_result <- unit_root_test(Gold_ts, "Gold")
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression none


Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
     Min       1Q   Median       3Q      Max 
-138.655  -26.586   -2.016   22.726  206.439 

Coefficients:
           Estimate Std. Error t value Pr(>|t|)
z.lag.1    0.003218   0.003322   0.969    0.334
z.diff.lag 0.122451   0.079603   1.538    0.126

Residual standard error: 52.57 on 157 degrees of freedom
Multiple R-squared:  0.02351,	Adjusted R-squared:  0.01107 
F-statistic:  1.89 on 2 and 157 DF,  p-value: 0.1545


Value of test-statistic is: 0.9687 

Critical values for test statistics: 
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62


##################################
# Phillips-Perron Unit Root Test #
##################################

Test regression with intercept


Call:
lm(formula = y ~ y.l1)

Residuals:
     Min       1Q   Median       3Q      Max 
-146.558  -28.725   -7.533   20.322  209.729 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 19.85676   13.11322   1.514    0.132    
y.l1         0.98898    0.01038  95.318   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 52.43 on 158 degrees of freedom
Multiple R-squared:  0.9829,    Adjusted R-squared:  0.9828
F-statistic:  9085 on 1 and 158 DF,  p-value: < 2.2e-16


Value of test-statistic, type: Z-tau  is: -1.0701

         aux. Z statistics
Z-tau-mu           1.5151

Critical values for Z statistics:
                     1pct      5pct     10pct
critical values -3.472136 -2.879539 -2.576262
```

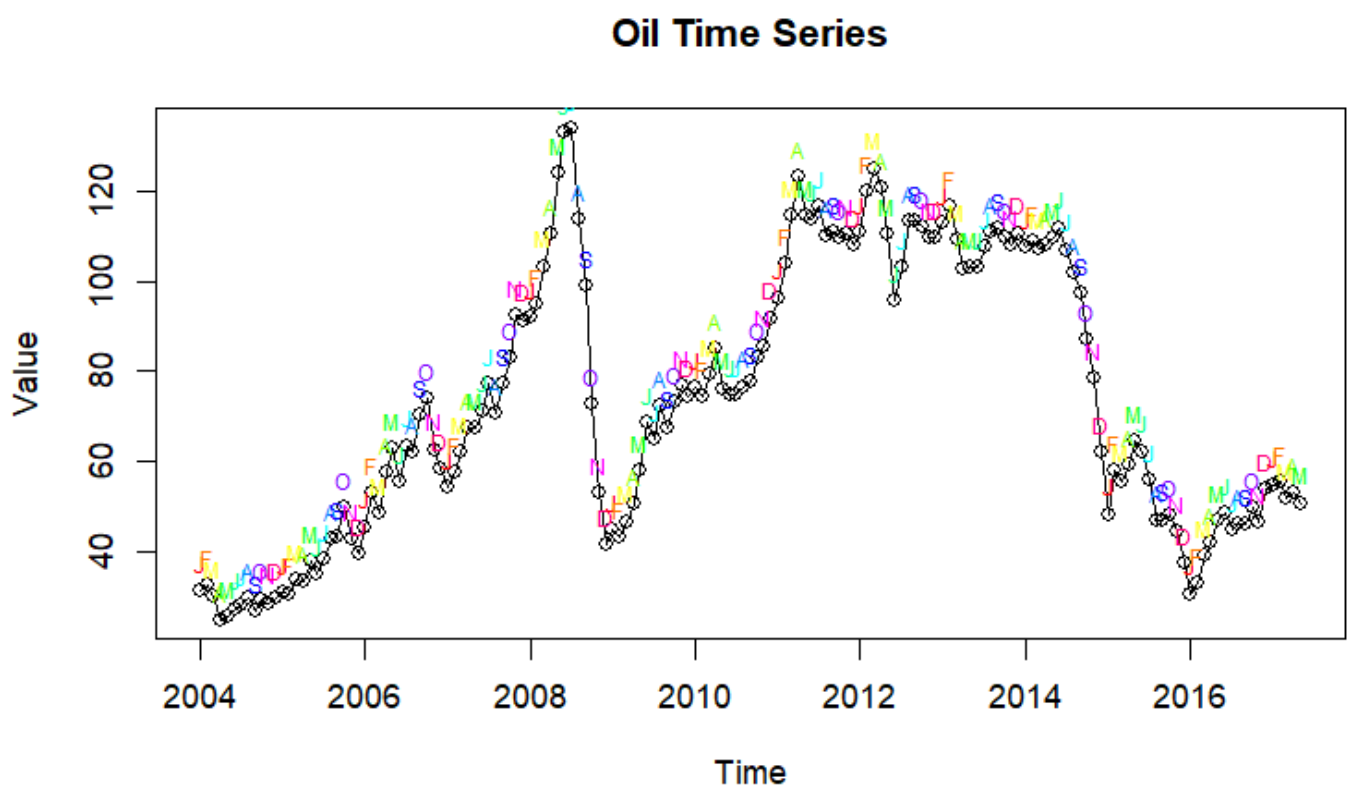
Oil variable

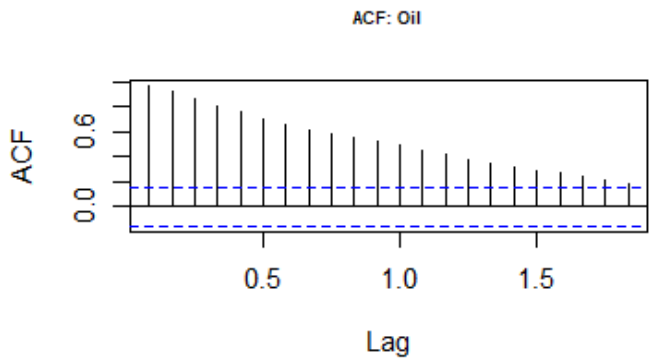
The Oil series shows a pronounced upward trend with increasing variance over time. Its ACF and PACF patterns indicate non-stationarity. Both the ADF and PP tests corroborate the visual evidence: the ADF test statistic is greater than the 5% critical value, and the PP test yields a similar result. Consequently, the null hypothesis of a unit root is not rejected.

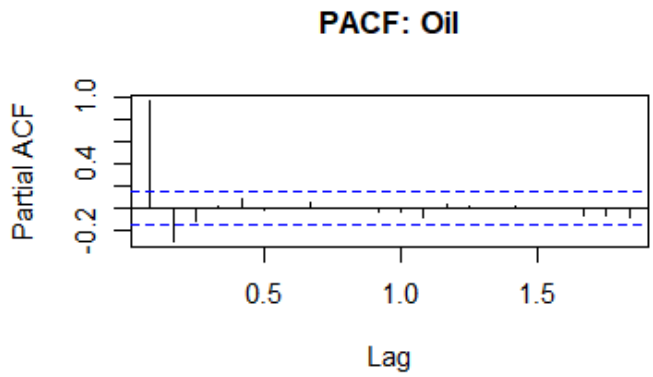
=> **Nonstationary**

Hide

```
# Running functions for Oil variable
plot_series(Oil_ts, "Oil")
```







Hide

```
Oil_result <- unit_root_test(Oil_ts, "Oil")
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression none


Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max 
-19.905  -3.562   1.204   3.988  15.202 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
z.lag.1    -0.004013   0.005806  -0.691     0.49    
z.diff.lag  0.400123   0.073236   5.463  1.8e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.839 on 157 degrees of freedom
Multiple R-squared:  0.1605,	Adjusted R-squared:  0.1498 
F-statistic: 15.01 on 2 and 157 DF,  p-value: 1.084e-06


Value of test-statistic is: -0.6913 

Critical values for test statistics: 
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62


##################################
# Phillips-Perron Unit Root Test #
##################################

Test regression with intercept


Call:
lm(formula = y ~ y.l1)

Residuals:
     Min       1Q   Median       3Q      Max 
-25.6715  -3.5936   0.4549   3.9455  14.3505 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.07981    1.32129   1.574    0.117    
y.l1         0.97347    0.01659  58.686   <2e-16 ***
```

```
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.302 on 158 degrees of freedom
Multiple R-squared:  0.9561,	Adjusted R-squared:  0.9559
F-statistic:  3444 on 1 and 158 DF,  p-value: < 2.2e-16


Value of test-statistic, type: Z-tau  is: -2.0125

         aux. Z statistics
Z-tau-mu           1.9323

Critical values for Z statistics:
                      1pct      5pct     10pct
critical values -3.472136 -2.879539 -2.576262
```

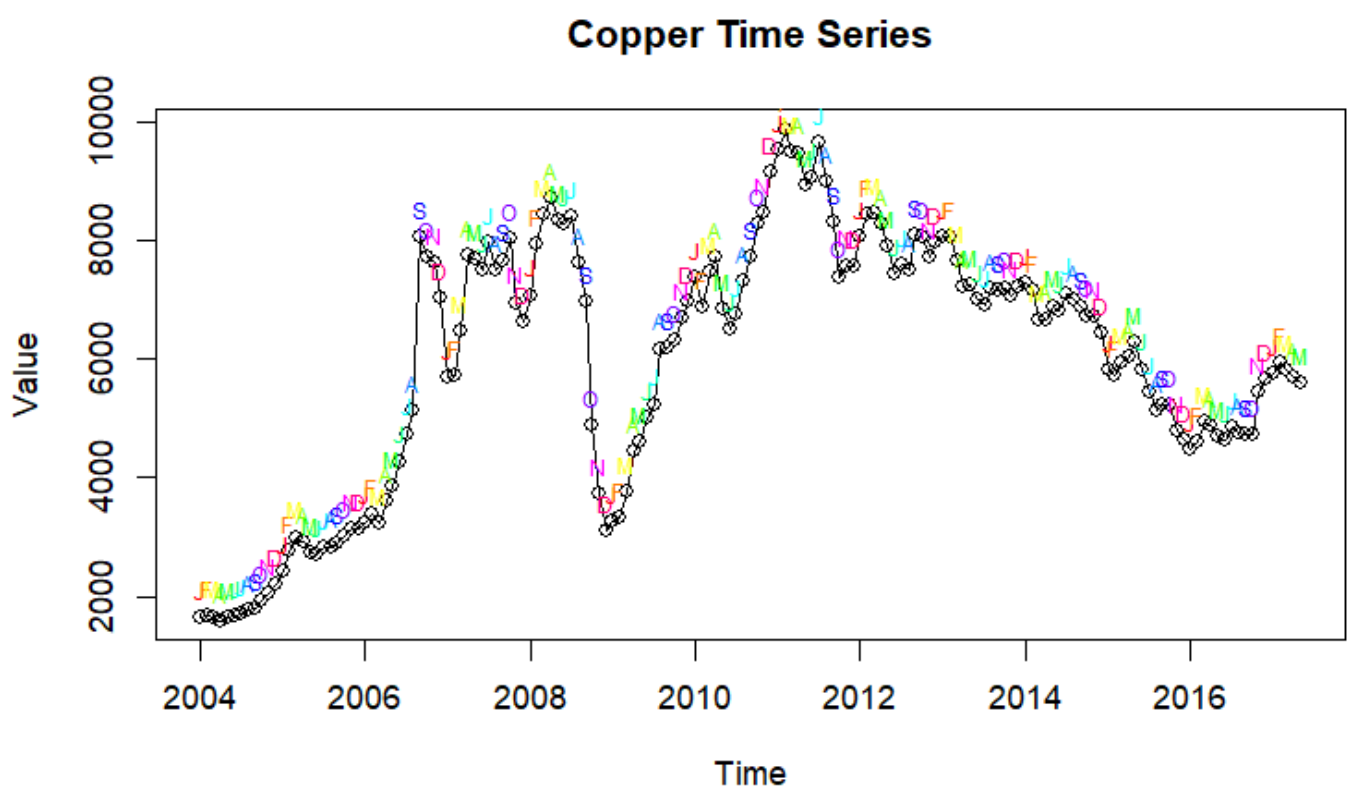
Copper variable

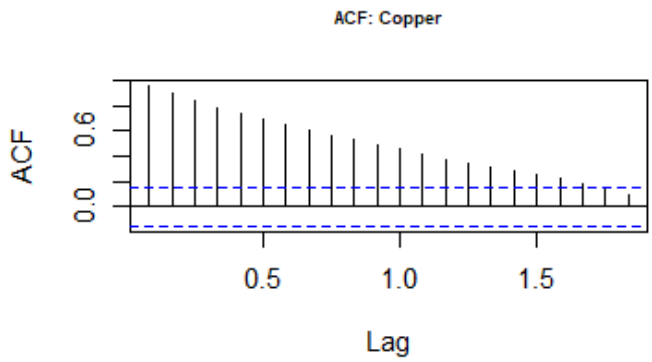
Finally, Copper prices exhibit a strong upward trend. The ACF shows a slow decay, and the PACF has a high first autocorrelation. The ADF test returns a statistic of -0.3909, which is greater than all critical values, and the PP test statistic also exceeds its critical value. Therefore, both tests fail to reject the null hypothesis.

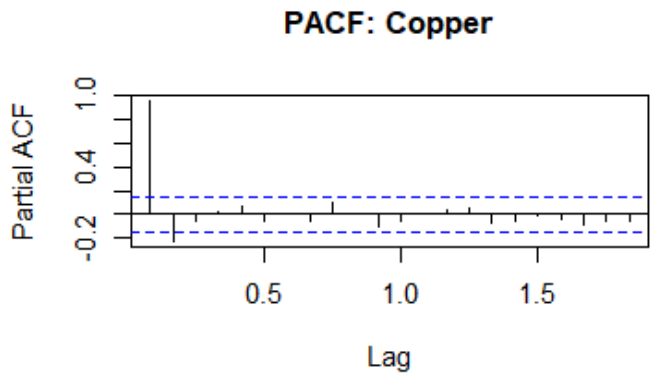
=> **Nonstationary**

Hide

```
# Running functions for Copper variable
plot_series(Copper_ts, "Copper")
```







Hide

```
Copper_result <- unit_root_test(Copper_ts, "Copper")
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression none


Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
     Min       1Q   Median       3Q      Max 
-1861.58  -164.87    32.72   248.08  2829.69 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
z.lag.1    -0.002256   0.005772  -0.391    0.696    
z.diff.lag  0.307564   0.076139   4.040 8.36e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 463.1 on 157 degrees of freedom
Multiple R-squared:  0.09422,	Adjusted R-squared:  0.08268 
F-statistic: 8.166 on 2 and 157 DF,  p-value: 0.0004229


Value of test-statistic is: -0.3909 

Critical values for test statistics: 
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62


##################################
# Phillips-Perron Unit Root Test #
##################################

Test regression with intercept


Call:
lm(formula = y ~ y.l1)

Residuals:
     Min       1Q   Median       3Q      Max 
-2066.88  -213.60   -21.83   225.07  2876.96 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 255.00356  112.61252   2.264   0.0249 *  
y.l1          0.96156    0.01771  54.301   <2e-16 ***
```

```
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 477.4 on 158 degrees of freedom
Multiple R-squared:  0.9491,    Adjusted R-squared:  0.9488
F-statistic:  2949 on 1 and 158 DF,  p-value: < 2.2e-16


Value of test-statistic, type: Z-tau  is: -2.3641

         aux. Z statistics
Z-tau-mu             2.406

Critical values for Z statistics:
                     1pct      5pct     10pct
critical values -3.472136 -2.879539 -2.576262
```

Exploration of components of the time series data

Stationarity

- All four variables were found to be non-stationary. To address this:
  - At first, The Box-Cox transformation was applied to identify variables requiring transformation. Oil was the only variable transformed ($\lambda < 1$).
  - First-order differencing was then applied to all variables. After differencing, all series passed the ADF test, confirming stationarity.

Hide

```
#Calculating lambda function
get_lambda <- function(ts_name, name) {
  lambda = BoxCox.lambda(ts_name)
  cat("lambda of",name, lambda, "\n")
}

#get lambda
get_lambda(ASX_ts, "ASX")
```

```
lambda of ASX 1.999924
```

Hide

```
get_lambda(Gold_ts, "Gold")
```

```
lambda of Gold 0.976695
```

Hide

```
get_lambda(Oil_ts, "Oil")
```

```
lambda of Oil -0.8304136
```

Hide

```
get_lambda(Copper_ts, "Copper")
```

```
lambda of Copper 0.9336783
```

Hide

```
#Difference variables
get_diff_adf <- function(ts_name, name) {
    ts_name <- diff(ts_name)       # difference
    cat("ADF test for", name)
    print(adf.test(ts_name))               # run ADF test
}

# Running function
get_diff_adf(ASX_ts, "ASX")
```

```
ADF test for ASX
```

```
Warning :p-value smaller than printed p-value
```

```

    Augmented Dickey-Fuller Test

data:  ts_name
Dickey-Fuller = -4.5543, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

Hide

```
get_diff_adf(Gold_ts, "Gold")
```

```
ADF test for Gold
```

```
Warning :p-value smaller than printed p-value
```

```

    Augmented Dickey-Fuller Test

data:  ts_name
Dickey-Fuller = -5.8718, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

Hide

```
get_diff_adf(log(Oil_ts), "Oil")
```

```
ADF test for Oil
```

```
Warning :p-value smaller than printed p-value
```

```

    Augmented Dickey-Fuller Test

data:  ts_name
Dickey-Fuller = -5.539, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

Hide

```
get_diff_adf(Copper_ts, "Copper")
```

```
ADF test for Copper
```

```
Warning :p-value smaller than printed p-value
```

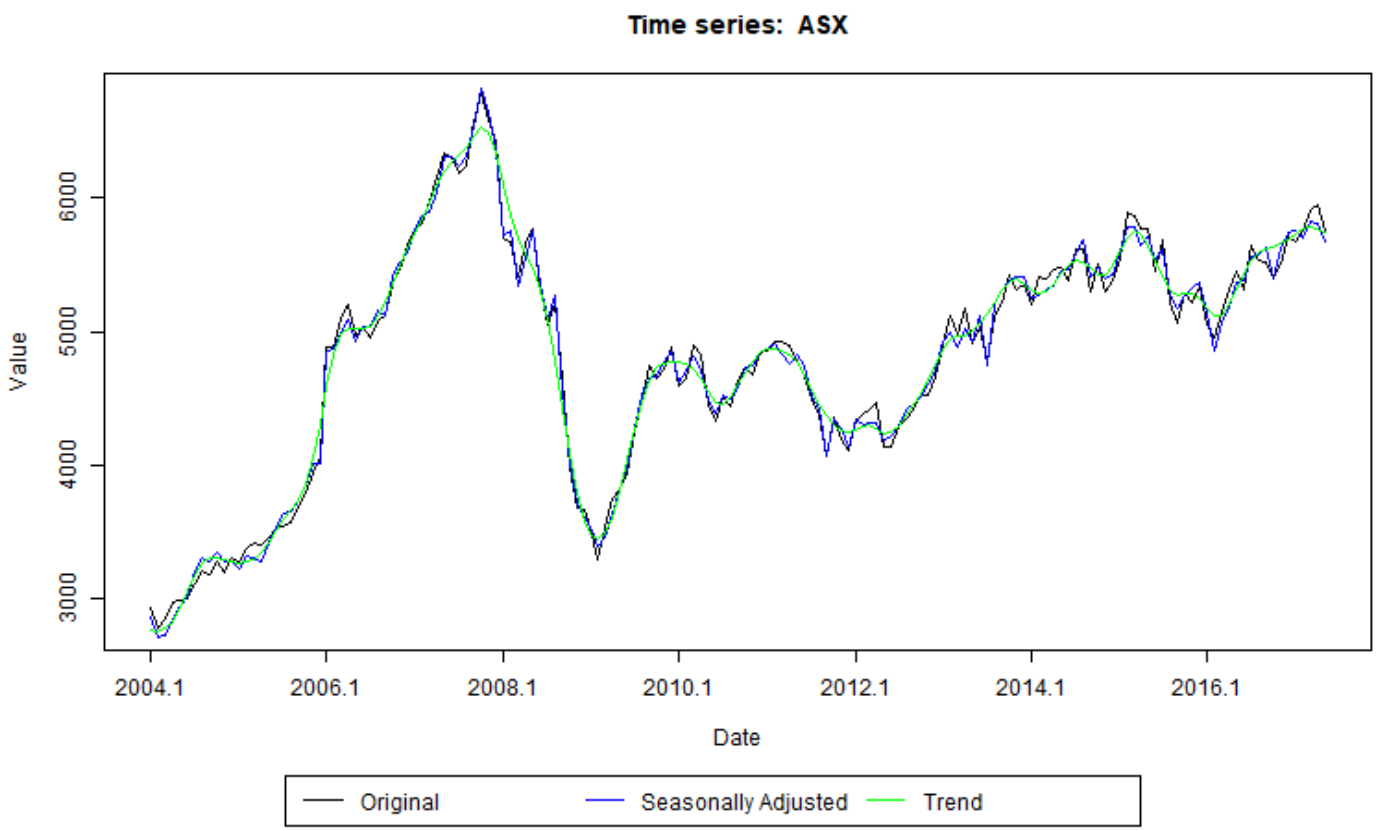
```

    Augmented Dickey-Fuller Test

data:  ts_name
Dickey-Fuller = -5.478, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary
```

Components of time series data

Hide

```
#Decomposition to analyse components of time series
decomp.x12 <- function(ts_name,name) {
  decompose_name <- x12(ts_name)
  plot(decompose_name,main=paste("Time series: ", name), sa=TRUE, trend=TRUE)
}

decomp.x12(ASX_ts, "ASX")
```

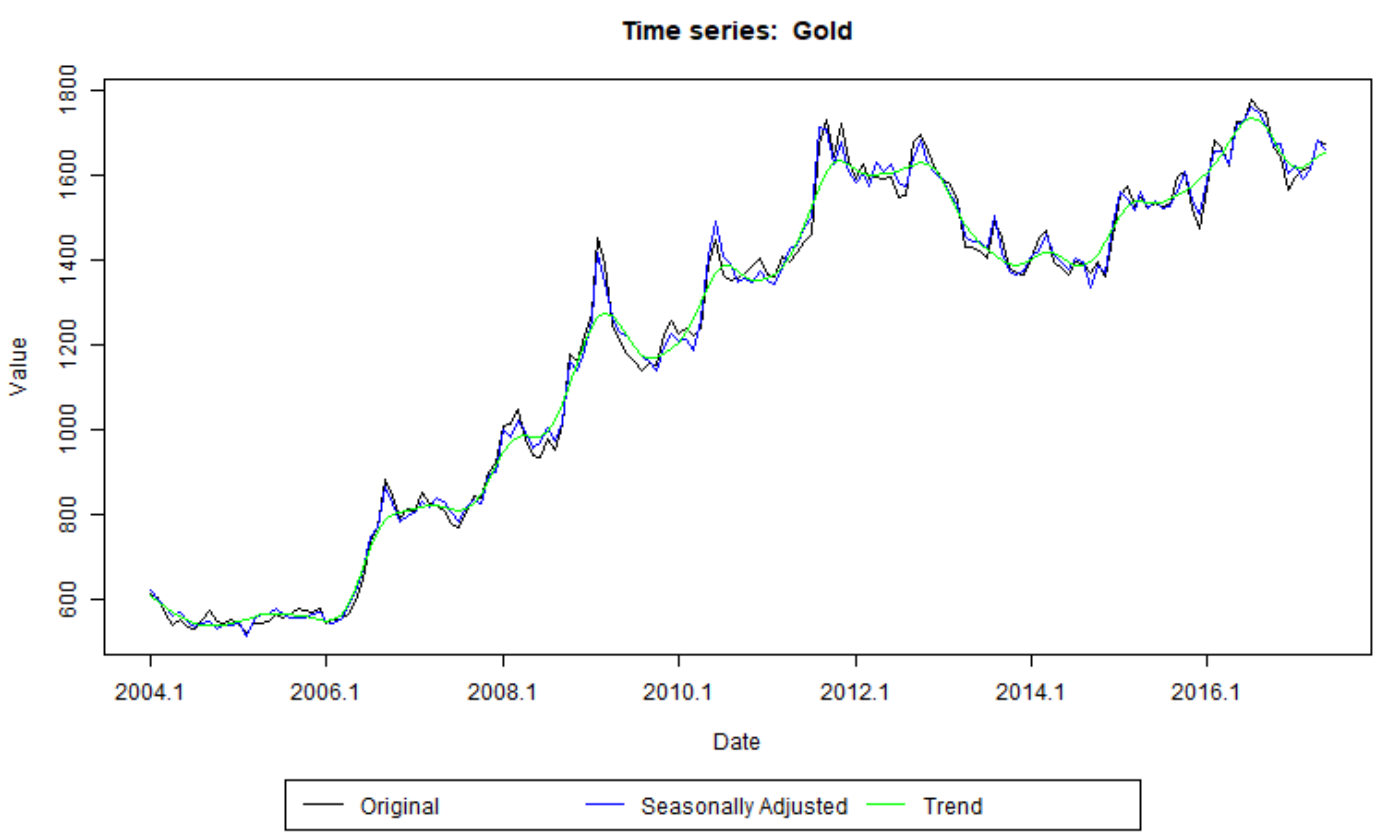
```
X-13ARIMA-SEATS Seasonal Adjustment Program
Version Number 1.1 Build 60
Execution began  082  0,    0  00.00.00

 Reading input spec file from Rout.spc
 Storing any program output into Rout.html
 Storing any program error messages into Rout_err.html
 Storing any diagnostics output into gra_Rout/Rout.udg


 NOTE: The X-13ARIMA-SEATS diagnostic file (.udg) has been stored
       in the directory specified by the graphics (-g) option.

 NOTE: The X-13ARIMA-SEATS diagnostic file (.udg) is generated
       since the graphics (-g) option was specified.

 WARNING: At least one visually significant trading day peak has been
          found in the estimated spectrum of the regARIMA residuals.

 WARNING: At least one visually significant trading day peak has been
          found in one or more of the estimated spectra.

 WARNING: Series should not be a candidate for seasonal adjustment
           because the spectrum of the original series (Table A1 or B1)
           has no visually significant seasonal peaks.

Execution complete for Rout.spc at  082  0,    0  00.00.00
```



Hide

```
decomp.x12(Gold_ts, "Gold")
```

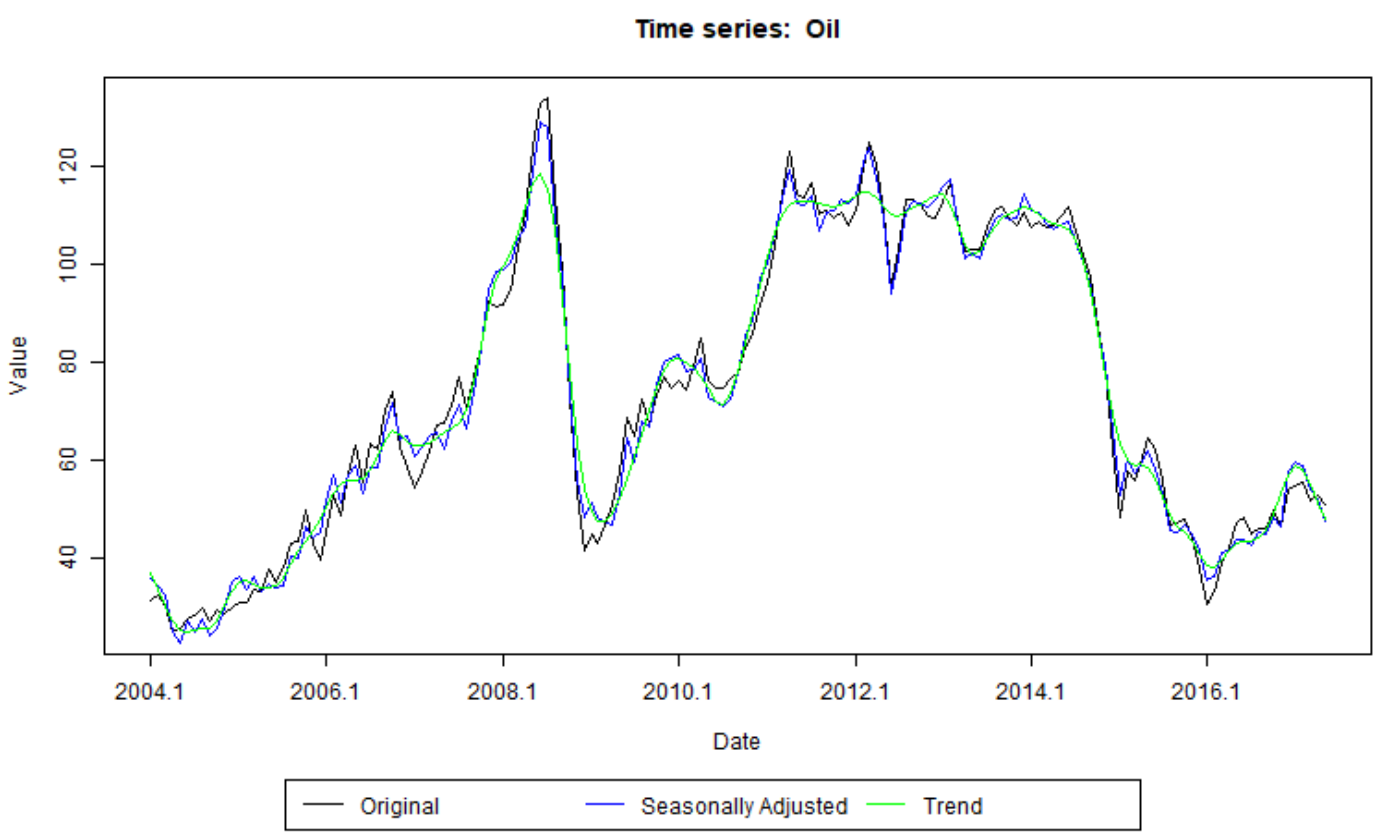
```
X-13ARIMA-SEATS Seasonal Adjustment Program
Version Number 1.1 Build 60
Execution began  082  0,    0  00.00.00

 Reading input spec file from Rout.spc
 Storing any program output into Rout.html
 Storing any program error messages into Rout_err.html
 Storing any diagnostics output into gra_Rout/Rout.udg


 NOTE: The X-13ARIMA-SEATS diagnostic file (.udg) has been stored
       in the directory specified by the graphics (-g) option.

 NOTE: The X-13ARIMA-SEATS diagnostic file (.udg) is generated
       since the graphics (-g) option was specified.

 WARNING: Visually significant seasonal and trading day peaks have
          been found in the estimated spectrum of the regARIMA residuals.

 WARNING: At least one visually significant trading day peak has been
          found in one or more of the estimated spectra.

Execution complete for Rout.spc at  082  0,    0  00.00.00
```



Hide

```
decomp.x12(Oil_ts, "Oil")
```

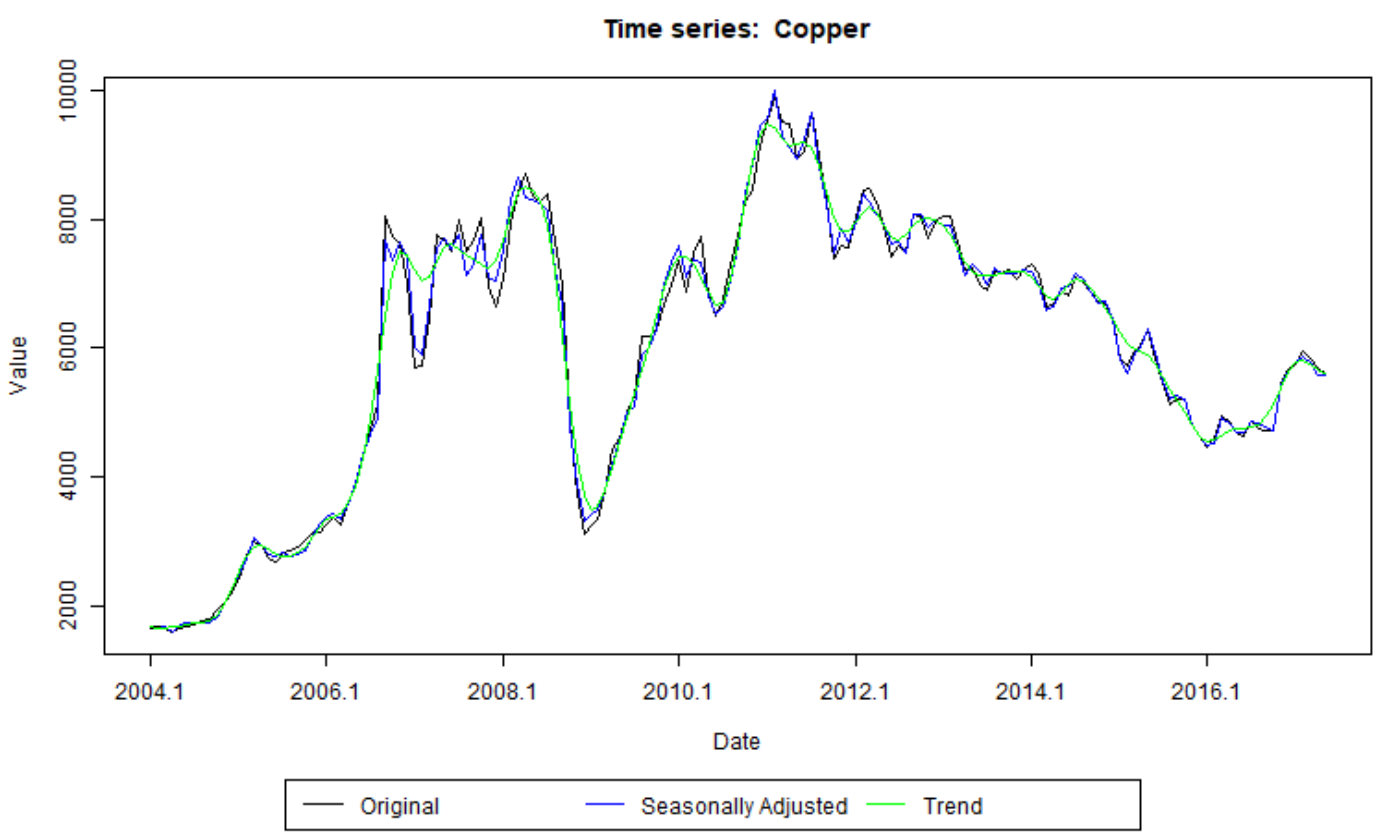
```
X-13ARIMA-SEATS Seasonal Adjustment Program
Version Number 1.1 Build 60
Execution began  082  0,    0  00.00.00

 Reading input spec file from Rout.spc
 Storing any program output into Rout.html
 Storing any program error messages into Rout_err.html
 Storing any diagnostics output into gra_Rout/Rout.udg


 NOTE: The X-13ARIMA-SEATS diagnostic file (.udg) has been stored
       in the directory specified by the graphics (-g) option.

 NOTE: The X-13ARIMA-SEATS diagnostic file (.udg) is generated
       since the graphics (-g) option was specified.

 WARNING: Series should not be a candidate for seasonal adjustment
          because the spectrum of the original series (Table A1 or B1)
          has no visually significant seasonal peaks.

Execution complete for Rout.spc at  082  0,    0  00.00.00
```



Hide

```
decomp.x12(Copper_ts, "Copper")
```

```
X-13ARIMA-SEATS Seasonal Adjustment Program
Version Number 1.1 Build 60
Execution began  082  0,    0  00.00.00

 Reading input spec file from Rout.spc
 Storing any program output into Rout.html
 Storing any program error messages into Rout_err.html
 Storing any diagnostics output into gra_Rout/Rout.udg


 NOTE: The X-13ARIMA-SEATS diagnostic file (.udg) has been stored
       in the directory specified by the graphics (-g) option.

 NOTE: The X-13ARIMA-SEATS diagnostic file (.udg) is generated
       since the graphics (-g) option was specified.

 WARNING: Series should not be a candidate for seasonal adjustment
          because the spectrum of the original series (Table A1 or B1)
          has no visually significant seasonal peaks.

Execution complete for Rout.spc at  082  0,    0  00.00.00
```



Assessment

- According to (Hyndman & Athanasopoulos, 2018), a time series can be decomposed into three main components: trend, seasonal, and cyclical. Among these, trend and seasonal patterns require careful identification and analysis.
- The plots indicate that the trend is the most dominant component across all four datasets. It is the primary driver of non-stationarity, meaning that the statistical properties of the series, such as the mean and variance, change over time. The persistent upward movement largely explains the overall increase in prices and highlights why simple statistical models would struggle to produce accurate forecasts without first addressing the trend.
- The seasonal component reflects regular fluctuations associated with calendar-based factors, such as months, quarters, or days of the week. However, both the raw time series plots and the decomposition results suggest that seasonality is absent in these datasets. Therefore, no seasonal adjustments are necessary in the analysis.
- Finally, the cyclical component represents long-term fluctuations that are not tied to a fixed period. In the ASX, Oil, and Copper series, cyclical patterns are evident, producing multi-year variations that significantly influence the behavior of the variables over time. These long-term cycles should be considered when analyzing the series.

Decision on distributed lag model for the ASX price index

Hide

```
# Correlation matrix
cor(data_ts)
```

```
                          ASX.price Gold.price Crude.Oil..Brent._USD.bbl Copper_USD.tonne
ASX.price                 1.0000000  0.3431908                 0.3290338        0.5617864
Gold.price                0.3431908  1.0000000                 0.4366382        0.5364213
Crude.Oil..Brent._USD.bbl 0.3290338  0.4366382                 1.0000000        0.8664296
Copper_USD.tonne          0.5617864  0.5364213                 0.8664296        1.0000000
```

- Copper and Oil are highly correlated, which could cause multicollinearity. Among the independent variables, Copper shows the highest correlation (0.56) with ASX. Therefore, Copper was chosen as the explanatory variable for subsequent modeling.

FINITE DLM

- The first model fitted was a finite distributed lag model, where the number of lags is pre-specified. A loop was conducted to identify the lag length that minimizes the AIC and BIC. The results indicated that a lag of 12 (q = 12) was optimal.

Hide

```
# loop for identifying number of lag for model
for (i in 1:12) {
  model1.1 = dlm(x=as.vector(data$Copper_USD.tonne),y=as.vector(data$ASX.price), q = i )
  cat ("q=", i, "AIC=", AIC(model1.1$model), "BIC=", BIC(model1.1$model), "\n") # printing AIC a
nd BIC
}
```

```
q= 1 AIC= 2574.488 BIC= 2586.789
q= 2 AIC= 2559.356 BIC= 2574.7
q= 3 AIC= 2544.155 BIC= 2562.531
q= 4 AIC= 2528.895 BIC= 2550.289
q= 5 AIC= 2513.265 BIC= 2537.664
q= 6 AIC= 2497.775 BIC= 2525.166
q= 7 AIC= 2481.988 BIC= 2512.357
q= 8 AIC= 2466.511 BIC= 2499.846
q= 9 AIC= 2451.016 BIC= 2487.302
q= 10 AIC= 2436.164 BIC= 2475.389
q= 11 AIC= 2420.623 BIC= 2462.771
q= 12 AIC= 2405.162 BIC= 2450.221
```

- This model explains only a small portion of the variation in the ASX index, as indicated by low R-squared and adjusted R-squared values. The residual error is relatively large, and although most lag coefficients are statistically significant, the model suffers from overfitting and weak explanatory power.

Hide

```
model1<-dlm(x=as.vector(data$Copper_USD.tonne),y=as.vector(data$ASX.price), q = 12) #apply model
summary(model1)
```

AIC	BIC
<dbl>	<dbl>
2405.162	2450.221

1 row

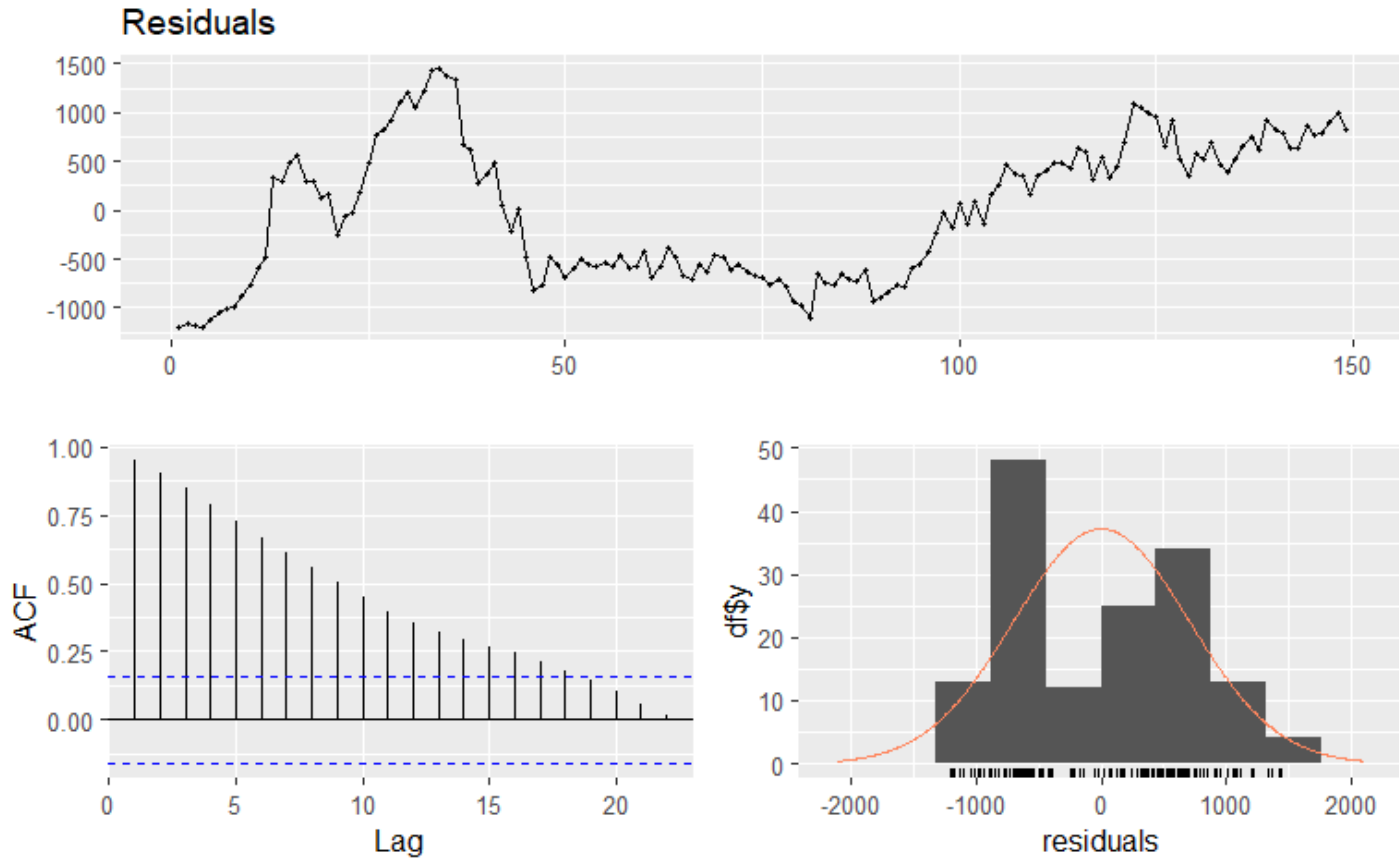
```
Call:
lm(formula = model.formula, data = design)

Residuals:
     Min       1Q   Median       3Q      Max 
-1196.59  -618.58    58.35   582.94  1441.94 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  4.117e+03  2.276e+02  18.087   <2e-16 ***
x.t          1.458e-01  1.369e-01   1.065    0.289    
x.1          2.374e-02  2.212e-01   0.107    0.915    
x.2          3.433e-02  2.239e-01   0.153    0.878    
x.3          2.489e-02  2.234e-01   0.111    0.911    
x.4          2.331e-02  2.225e-01   0.105    0.917    
x.5         -4.938e-02  2.209e-01  -0.224    0.823    
x.6          3.183e-02  2.207e-01   0.144    0.886    
x.7         -1.153e-02  2.218e-01  -0.052    0.959    
x.8         -2.387e-03  2.244e-01  -0.011    0.992    
x.9         -6.943e-02  2.252e-01  -0.308    0.758    
x.10        -2.573e-02  2.258e-01  -0.114    0.909    
x.11        -2.364e-02  2.232e-01  -0.106    0.916    
x.12         2.764e-02  1.357e-01   0.204    0.839    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 735.6 on 135 degrees of freedom
Multiple R-squared:  0.1692,	Adjusted R-squared:  0.08924 
F-statistic: 2.115 on 13 and 135 DF,  p-value: 0.01687

AIC and BIC values for the model:
```

- The residual diagnostics show significant autocorrelation and non-normality.

Hide

```
checkresiduals(model1$model)
```

```
	Breusch-Godfrey test for serial correlation of order up to 17

data:  Residuals
LM test = 137.27, df = 17, p-value < 2.2e-16
```



- Multicollinearity is also a major concern, with very high VIF values. Overall, this model is inadequate and unreliable for predictive purposes.

Hide

```
VIF.model1 <- vif(model1$model)
VIF.model1 >10
```

```
 x.t  x.1  x.2  x.3  x.4  x.5  x.6  x.7  x.8  x.9 x.10 x.11 x.12
TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
```

POLYNOMIAL DLM

- Despite this improvement, the residuals are still not randomly scattered around zero, and the ACF indicates high autocorrelation. Residuals are also not normally distributed, which can affect the reliability of statistical inferences.

Hide

```
model2 <- polyDlm(x=as.vector(data$Copper_USD.tonne), y=as.vector(data$ASX.price), q=12, k=2, sh
ow.beta = TRUE) #apply model
```

	Estimate <dbl>	Std. Error <dbl>	t value <dbl>	P(>|t|) <dbl>
beta.0	0.095900	0.02530	3.7900	2.26e-04
beta.1	0.068900	0.01530	4.5000	1.46e-05

	Estimate <dbl>	Std. Error <dbl>	t value <dbl>	P(>|t|) <dbl>
beta.2	0.045400	0.00867	5.2300	6.16e-07
beta.3	0.025300	0.00765	3.3000	1.23e-03
beta.4	0.008600	0.01020	0.8400	4.02e-01
beta.5	-0.004620	0.01250	-0.3700	7.12e-01
beta.6	-0.014400	0.01330	-1.0800	2.81e-01
beta.7	-0.020700	0.01250	-1.6600	9.91e-02
beta.8	-0.023600	0.01010	-2.3300	2.14e-02
beta.9	-0.023000	0.00718	-3.2100	1.67e-03

1-10 of 13 rows Previous **1** 2 Next

```
Estimates and t-tests for beta coefficients:
```

Hide

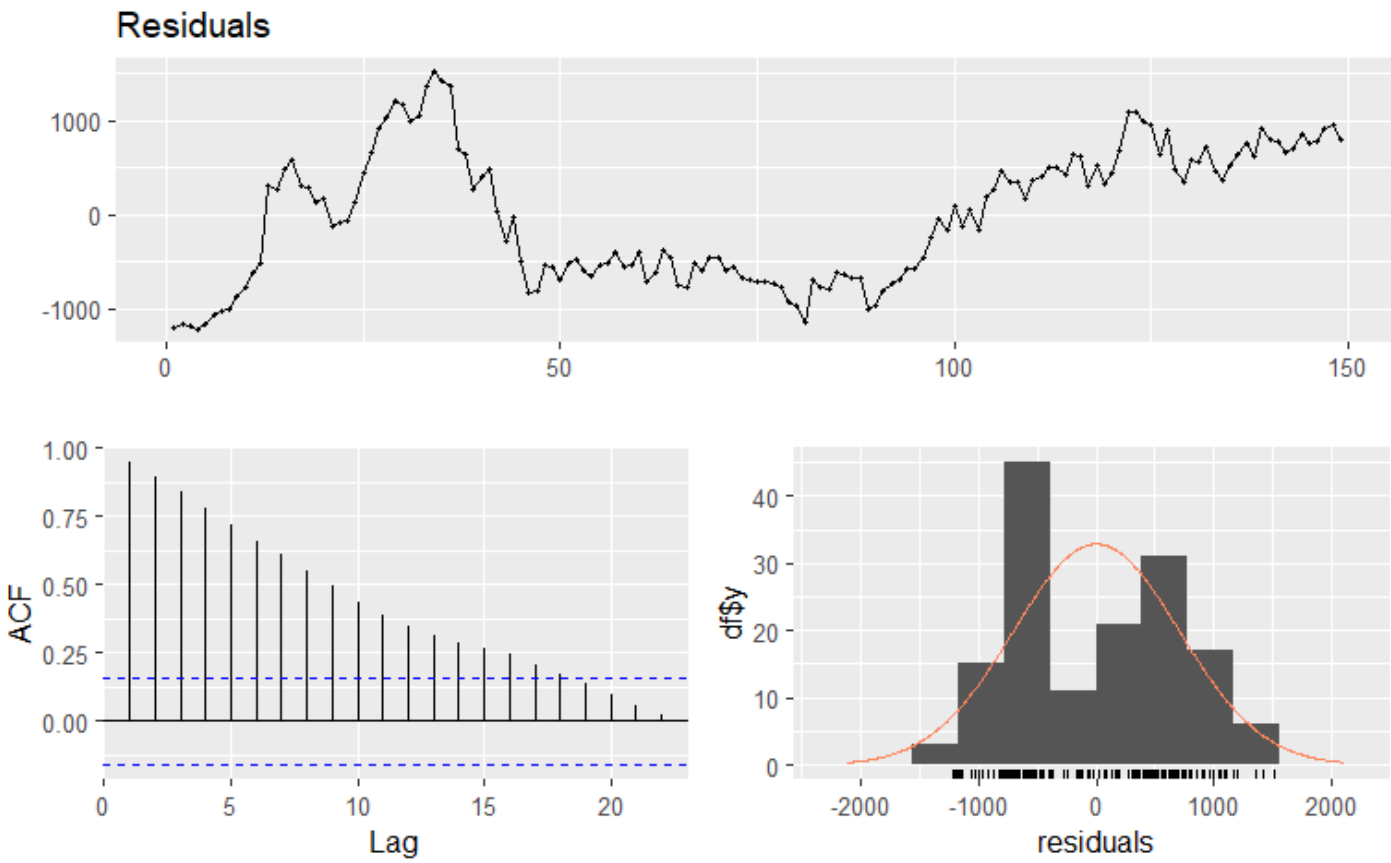
```
summary(model2)
```

```


Call:
"Y ~ (Intercept) + X.t"

Residuals:
     Min       1Q   Median       3Q      Max 
-1213.59  -610.66    26.99   585.36  1512.44 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  4.136e+03  2.163e+02  19.124  < 2e-16 ***
z.t0         9.592e-02  2.532e-02   3.788 0.000222 ***
z.t1        -2.872e-02  1.190e-02  -2.414 0.017022 *  
z.t2         1.723e-03  9.736e-04   1.770 0.078906 .  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 710.9 on 145 degrees of freedom
Multiple R-squared:  0.1666,	Adjusted R-squared:  0.1494 
F-statistic: 9.664 on 3 and 145 DF,  p-value: 7.422e-06
```

Hide

```
checkresiduals(model2$model)
```

```
    Breusch-Godfrey test for serial correlation of order up to 10

data:  Residuals
LM test = 135.33, df = 10, p-value < 2.2e-16
```



Hide

```
#sorting AIC of existence models
aic.sort <- AIC(model1$model, model2$model)
sortScore(aic.sort,"aic")
```

	df <dbl>	AIC <dbl>
model2$model	5	2385.629
model1$model	15	2405.162

2 rows

- The model has a lower AIC compared to the previous model, but multicollinearity remains a problem, as predictors are highly correlated.

Hide

```
VIF.model2 <- vif(model2$model)
VIF.model2 >10
```

```
z.t0 z.t1 z.t2
TRUE TRUE TRUE
```

KOYCK Model

- The Koyck model provides a strong overall fit, explaining 94.85% of the variance in the dependent variable with a residual error of 202.9. The model is statistically significant overall (Wald test p < 2.2e-16).
- Residual diagnostics indicate no significant autocorrelation (Ljung-Box p = 0.7953), and residuals are mostly centered around zero with no obvious trend, although some spikes and fat tails are observed.
- The model's geometric lag coefficient is small, suggesting a strong dynamic dependence of the ASX index on its past values, while the immediate effect of the independent variable is limited.

Hide

```
model3 <- koyckDlm(x=as.vector(data$Copper_USD.tonne), y=as.vector(data$ASX.price)) #apply model
summary(model3)
```

	alpha <dbl>	beta <dbl>	phi <dbl>
Geometric coefficients:	6672.885	-0.005863623	0.9716211

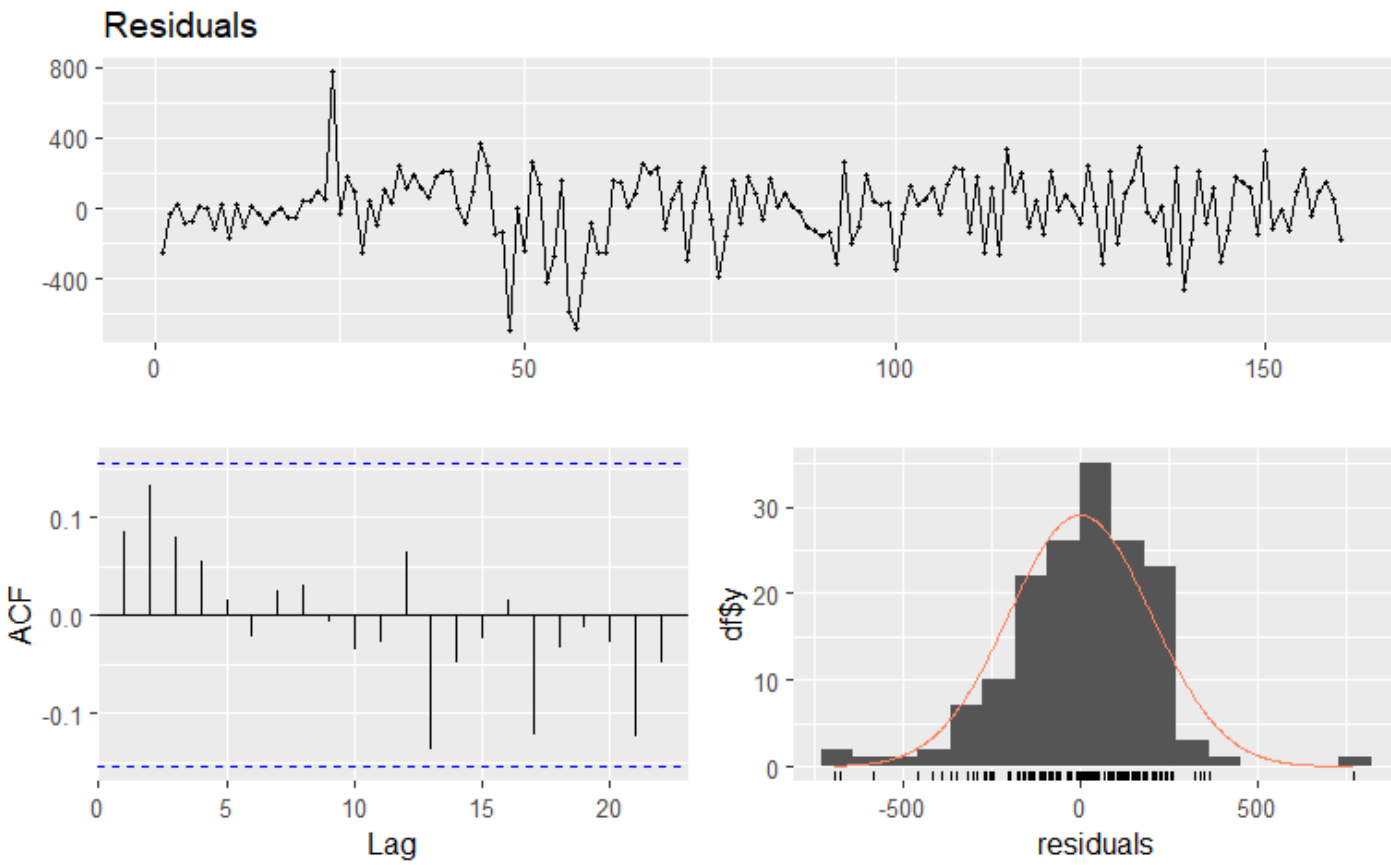
1 row

```
Call:
"Y ~ (Intercept) + Y.1 + X.t"

Residuals:
    Min      1Q  Median      3Q     Max 
-689.64 -108.62   12.78  140.20  771.79 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) 189.368812  87.644648   2.161   0.0322 *  
Y.1           0.971621   0.021895  44.376   <2e-16 ***
X.t          -0.005864   0.009517  -0.616   0.5387    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 201.9 on 157 degrees of freedom
Multiple R-Squared: 0.9485, Adjusted R-squared: 0.9479 
Wald test:  1448 on 2 and 157 DF,  p-value: < 2.2e-16

Diagnostic tests:
NULL
```

Hide

```
checkresiduals(model3$model)
```

```
    Ljung-Box test

data:  Residuals
Q* = 6.2327, df = 10, p-value = 0.7953

Model df: 0.   Total lags used: 10
```



- Importantly, the model does not show evidence of harmful multicollinearity, and parameter estimates are stable and reliable.

Hide

```
VIF.model3 <- vif(model3$model)
VIF.model3 > 10
```

```
  Y.1   X.t
FALSE FALSE
```

Autoregressive DLM

- An autoregressive distributed lag model was estimated using a loop to determine the optimal lag length based on AIC and BIC. The best model was identified with p = 5 and q = 1.

Hide

```
# Loop for identifying lag of dependent and independent variable
for (i in 1:5){
  for(j in 1:5){
    model4.1 = ardlDlm(x=as.vector(data$Copper_USD.tonne), y=as.vector(data$ASX.price), p = i ,
q = j )
    cat("p = ", i, "q = ", j, "AIC = ", AIC(model4.1$model), "BIC = ", BIC(model4.1$model),"\n")
  }
}
```

```
p =  1 q =  1 AIC =  2147.741 BIC =  2163.116
p =  1 q =  2 AIC =  2135.4 BIC =  2153.813
p =  1 q =  3 AIC =  2121.12 BIC =  2142.558
p =  1 q =  4 AIC =  2109.759 BIC =  2134.209
p =  1 q =  5 AIC =  2099.056 BIC =  2126.505
p =  2 q =  1 AIC =  2130.043 BIC =  2148.456
p =  2 q =  2 AIC =  2132.038 BIC =  2153.52
p =  2 q =  3 AIC =  2119.241 BIC =  2143.741
p =  2 q =  4 AIC =  2107.649 BIC =  2135.155
p =  2 q =  5 AIC =  2097.021 BIC =  2127.52
p =  3 q =  1 AIC =  2117.307 BIC =  2138.745
p =  3 q =  2 AIC =  2119.247 BIC =  2143.748
p =  3 q =  3 AIC =  2119.696 BIC =  2147.259
p =  3 q =  4 AIC =  2108.537 BIC =  2139.1
p =  3 q =  5 AIC =  2097.832 BIC =  2131.38
p =  4 q =  1 AIC =  2105.916 BIC =  2130.366
p =  4 q =  2 AIC =  2107.774 BIC =  2135.28
p =  4 q =  3 AIC =  2108.608 BIC =  2139.17
p =  4 q =  4 AIC =  2110.085 BIC =  2143.704
p =  4 q =  5 AIC =  2099.454 BIC =  2136.052
p =  5 q =  1 AIC =  2095.118 BIC =  2122.566
p =  5 q =  2 AIC =  2096.96 BIC =  2127.459
p =  5 q =  3 AIC =  2097.887 BIC =  2131.436
p =  5 q =  4 AIC =  2099.497 BIC =  2136.095
p =  5 q =  5 AIC =  2101.419 BIC =  2141.067
```

- This model explains about 95% of the variance in the dependent variable, indicating an excellent fit. Both the intercept and the contemporaneous value of the independent variable are statistically significant at the 5% level.
- Residual diagnostics show a standard error of 193.4, and the Breusch–Godfrey test (p = 0.93) confirms the absence of serial correlation, indicating well-behaved residuals.
- This model has lower AIC (2095.1) compared to the FDL (2405.2) and PDL (2385.6) models, demonstrating better efficiency and parsimony despite including multiple lags.

Hide

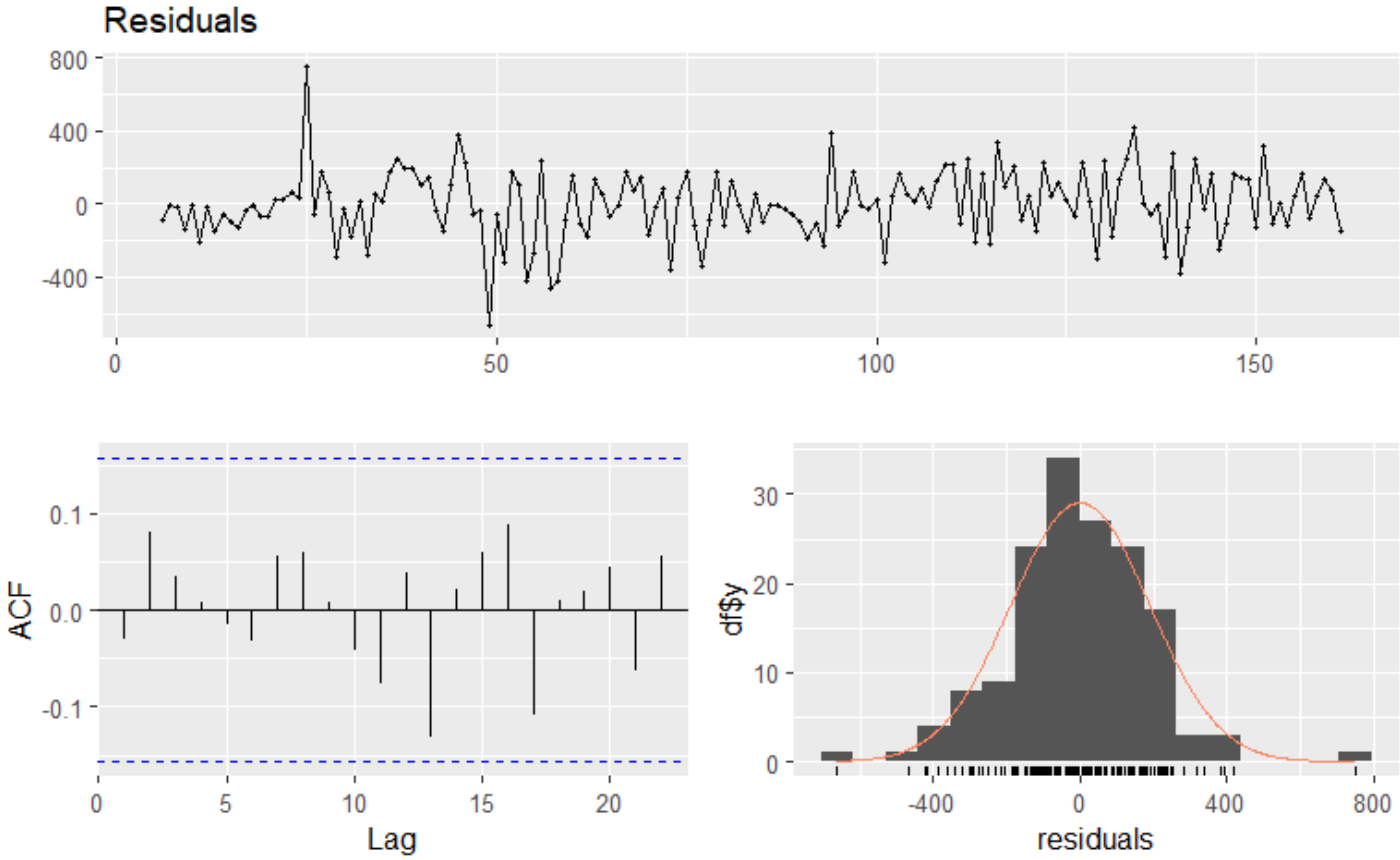
```
model4_51 <-ardlDlm(x=as.vector(data$Copper_USD.tonne), y=as.vector(data$ASX.price), p=5, q=1) #
apply model
summary(model4_51)
```

```
Time series regression with "ts" data:
Start = 6, End = 161

Call:
dynlm(formula = as.formula(model.text), data = data, start = 1)

Residuals:
    Min      1Q  Median      3Q     Max 
-663.01 -104.28   -6.52  134.04  748.83 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 223.07851   91.31917   2.443   0.0157 *  
X.t           0.08180    0.03445   2.374   0.0189 *  
X.1          -0.02382    0.05497  -0.433   0.6654    
X.2          -0.02426    0.05554  -0.437   0.6629    
X.3          -0.01059    0.05554  -0.191   0.8490    
X.4          -0.01423    0.05498  -0.259   0.7962    
X.5          -0.01564    0.03396  -0.461   0.6457    
Y.1           0.96496    0.02137  45.160   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 193.4 on 148 degrees of freedom
Multiple R-squared:  0.9492,	Adjusted R-squared:  0.9468 
F-statistic: 395.2 on 7 and 148 DF,  p-value: < 2.2e-16
```

Hide

```
checkresiduals(model4_51$model)
```

```
	Breusch-Godfrey test for serial correlation of order up to 11

data:  Residuals
LM test = 5.0114, df = 11, p-value = 0.9306
```



Hide

```
# sorting model
aic.sort <- AIC(model1$model, model2$model, model4_51$model)
```

```
Warning :models are not all fitted to the same number of observations
```

Hide

```
sortScore(aic.sort,"aic")
```

	df <dbl>	AIC <dbl>
model4_51$model	9	2095.118
model2$model	5	2385.629
model1$model	15	2405.162

3 rows

- However, the model shows signs of multicollinearity, which could affect the stability of some coefficient estimates.

Hide

```
VIF.model4_51 <- vif(model4_51$model)
VIF.model4_51 > 10
```

```
     X.t L(X.t, 1) L(X.t, 2) L(X.t, 3) L(X.t, 4) L(X.t, 5) L(y.t, 1)
    TRUE      TRUE      TRUE      TRUE      TRUE      TRUE     FALSE
```

Discussion

- Comparing all models, the Koyck and ARDL models perform best, explaining over 94% of ASX variation and showing low residual errors.
    - Koyck Model: Simple, dynamic, and reliable, with minimal multicollinearity. Captures the ASX's dependence on past values but shows limited immediate effect from predictors.
    - RDL Model: Explains slightly more variance and allows flexible lag structures, but multicollinearity may affect coefficient stability.
- Considering simplicity, reliability, and dynamic representation, the Koyck model is the most suitable. It balances strong explanatory power, parsimony, and stability, making it the preferred choice for modeling the ASX index.

Conclusion

This report analyzed the ASX All Ordinaries (ASX) Price Index using time series methods to identify a reliable model for forecasting. The dataset was first cleaned and transformed into appropriate time series objects. Non-stationarity was assessed through visual inspection and formal unit root tests (ADF and PP), and first-order differencing was applied to all variables to achieve stationarity. Trend was identified as the dominant component, while seasonality was absent. Correlation analysis led to the selection of Copper as the key explanatory variable.

Four models were compared: finite distributed lag (FDL), polynomial distributed lag (PDL), Koyck, and autoregressive distributed lag (ARDL). The Koyck model is preferred due to its simplicity, stability, and minimal multicollinearity, while the ARDL model offers flexibility but may be affected by correlated predictors. Overall, the Koyck is the most suitable distributed lag model. ## References Chand, P. (2023, May 9). *Stationary V. Non-Stationary Time Series*. Medium. https://medium.com/@prateekchand208/stationary-v-non-stationary-time-series-41dfa4514a8 (https://medium.com/@prateekchand208/stationary-v-non-stationary-time-series-41dfa4514a8)

Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting : Principles and Practice* (2nd ed.). Otexts. https://otexts.com/fpp2/ (https://otexts.com/fpp2/)